

Automation or Rising Wealth?

A within-dataset horse race between technological push and demand pull in the growth of US low-skill service employment, 1980–2005

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Dataset: Autor & Dorn (2013), American Economic Review — 722 US Commuting Zones, 1980–2005.

1. Research question and contribution

Between 1980 and 2005 the US labour market polarised into a U-shape: employment and wages grew at the top and bottom of the occupational skill distribution and stagnated in the middle. The most striking part of the lower tail is the rise of *low-skill service occupations* (food service, cleaning, security, personal care, child care). Among noncollege workers the service share of hours worked rose by more than 50 percent, and real service wages grew faster than in any other low-skill occupation group (Autor & Dorn, 2013). Two mechanisms have been proposed for this rise, and the literature has tested them *separately*:

- **Story A — Technological push (Autor & Dorn, 2013).** The falling cost of computing automates routine, codifiable tasks, which are concentrated in middle-skill clerical and production occupations. Noncollege workers displaced from those jobs reallocate into manual, hard-to-automate service work. *Prediction:* commuting zones (CZs) more specialised in routine tasks in 1980 should see larger subsequent growth of noncollege service employment.
- **Story B — Demand pull (Moretti, 2010).** Growth of well-paid tradable-sector jobs raises local incomes; high earners spend locally on in-person services, generating a local employment multiplier. *Prediction:* CZs with faster growth in top/college wages should see larger service-employment growth.

Research question. When both mechanisms are placed in the *same* specification on the *same* data, which one accounts for the cross-CZ growth of low-skill service employment over 1980–2005?

Contribution. This is a **replicate-and-extend** exercise, not a re-visualisation. We (i) *replicate* Autor & Dorn’s headline cross-CZ relationship between the 1980 routine employment share and subsequent noncollege service-employment growth, and (ii) *extend* it by introducing a Moretti-style demand-pull regressor — local top-/college-wage growth and college labour-supply growth — into the same instrumented specification. This is a direct *horse race* that neither paper runs: Autor & Dorn argue technology dominates but never include a demand-pull regressor, while Moretti documents local multipliers in a different research design. The value added is *adjudication* on identical data, with explicit predicted signs (β on routine share > 0 under Story A; β on wage growth > 0 under Story B) and a clear statement of which survives conditioning and instrumentation.

Two hypotheses, opposite empirical fingerprints

Hypothesis	Mechanism	Key regressor	Predicted sign
A — Technological push	Automation of routine tasks pushes displaced noncollege workers into services	RSH_{1980}	$\beta_1 > 0$, robust to demand controls
B — Demand pull	Rising local high incomes raise demand for in-person services (local multiplier)	$\Delta W90 / \Delta Colls$	$\beta_2, \beta_3 > 0$; β_1 should attenuate

The two stories are not mutually exclusive, but they imply different coefficient patterns in the same regression; the horse race in §4 lets the data choose between them.

2. Literature context

- **Autor, Levy & Murnane (2003), QJE — conceptual & measurement foundation.** Introduces the task framework in which computers substitute for routine tasks and complement abstract tasks, hitting the middle of the skill distribution hardest. The Routine Task Intensity (RTI) index we use derives directly from this paper.
- **Autor & Dorn (2013), AER — main reference and data source.** Builds a spatial-equilibrium model of routine-task-replacing technical change and tests it across 722 CZs; the instrumented routine-share regression is the relationship we replicate, and the replication package is our dataset.

- **Moretti (2010), AER — demand-side counterpoint.** “Local multipliers”: each additional tradable (especially high-tech) job creates several local nontradable/service jobs through local demand. This is the mechanism our extension regressor proxies, and the direct contrast to the technological-push story.
- **Mazzolari & Ragusa (2013), REStat — bridging channel.** High-skill consumption and rising returns to skill spill over into demand for low-skill services, motivating the college labour-supply / substitution proxy (which Autor & Dorn include as a control).
- **Goos, Manning & Salomons (2014), AER — external corroboration.** Cross-national evidence that routine task content, rather than offshorability, is the primary driver of polarisation — support for treating routine intensity as the leading candidate mechanism.

Where this proposal sits. Autor & Dorn and Moretti are usually read as complementary descriptions of the same era rather than as rival explanations to be weighed against one another. Autor & Dorn estimate the technology channel without a demand-pull regressor; Moretti estimates a local multiplier without the routine-task machinery and on a different sample. Because both mechanisms predict more service employment in the same booming local labour markets, the descriptive evidence each side cites is consistent with the other. Our proposal occupies the gap between them: we hold the dataset, the geography and the time window fixed, and let the two mechanisms compete inside a single regression. The aim is not to declare one theory wrong, but to measure how much of the cross-CZ variation each accounts for once the other is present.

3. Data

Source and provenance. The official replication package of Autor & Dorn (2013), deposited in the AEA Data and Code Repository / openICPSR, **project 112652** (DOI 10.3886/E112652V1); the CZ panel is the *workfile2012.dta* file from Dorn’s “P2” package. The data are public, peer-reviewed and fully documented. (This corrects the project number shown on our presentation slide, which read 112083.)

Unit of observation and coverage. US Commuting Zones (Tolbert & Sizer, 1996) — 722 local labour markets covering the entire mainland US, metropolitan and rural. Time coverage spans the Census years 1950, 1970, 1980, 1990, 2000 and the 2005 ACS; the main analysis window is 1980–2005, organised as three stacked decadal differences (1980–90, 1990–2000, 2000–05), giving $722 \times 3 = 2,166$ CZ-period observations.

Three files, three roles

- **workfile2012.dta — the main CZ panel** (3,610 rows = 722 CZs $\times$ 5 census years; 111 variables). Key variables: *l_sh_routine33a* (start-of-period routine employment share, *RSH* — our Story-A regressor; population-weighted mean 0.33 in 1980); *d_shocc1_service_nc* (decadal change in the noncollege service-employment share — the dependent variable); *R33a_50_** (the 1950 industry-mix instrument); *d_ln90wlsftfy* (change in the 90th-percentile full-time weekly wage — Story-B income proxy); *d_avgls2080_edu_coll* (change in college-graduate labour supply — substitution proxy); *l_task_std_offshore* (offshorability); plus controls (relative skill supply, immigrant share, manufacturing share, unemployment, female employment, elderly share, minimum-wage bite).
- **RTI-bypctile-1980.dta — the automation profile** (100 rows, one per 1980 occupational-skill percentile). *perc_R33a* gives the routine occupation share by skill percentile — the inverse-U profile that locates routine jobs in the middle of the distribution, and the basis for our high/low-routine split.
- **dwg-byperc-1980-2005-czall.dta — the polarisation evidence** (100 rows by 1980 wage percentile). Observed and counterfactual employment- and wage-growth by percentile across all CZs — the raw material for the U-shape figure.

Key variables (workfile2012.dta)

Variable	Description	Role
<i>l_sh_routine33a</i>	Start-of-period routine employment share (RSH)	Story A regressor
<i>d_shocc1_service_nc</i>	Δ noncollege service-employment share (decadal)	Dependent var.
<i>R33a_50_*</i>	Routine share predicted by 1950 industry mix	Instrument
<i>d_ln90wlsftfy</i>	Δ 90th-percentile weekly wage (income effect)	Story B regressor
<i>d_avgls2080_edu_coll</i>	Δ college-graduate labour supply (substitution)	Story B regressor

<i>l_task_std_offshore</i>	Offshorability index (FFL 2011)	Control
<i>timepwt48</i>	Start-of-period CZ population weight	Weight

Variable construction and weighting

The routine share rests on a chain of well-documented steps that we inherit unchanged from the package. Task content (abstract, routine, manual) is taken from the US Department of Labor’s *Dictionary of Occupational Titles* (1977) and mapped to a balanced panel of Census occupations; the occupation-level Routine Task Intensity index is $RTI = \ln(\text{routine}) - \ln(\text{manual}) - \ln(\text{abstract})$. An occupation is flagged “routine-intensive” if its 1980 RTI lies in the top employment-weighted third, and a CZ’s routine share RSH is the local employment share of those occupations. The dependent variable is built from the same occupation panel as the share of noncollege hours worked in service occupations (police and fire-fighters excluded, following Autor & Dorn). Every statistic is weighted by the product of the Census sampling weight, hours of labour supply, and the start-of-period CZ population share (“timepwt48”), so that large labour markets receive proportionate influence. We will take these constructions as given and audit only the pieces our argument leans on — the 1980 mean of RSH (0.33), its 80/20 range (0.072), and the first-stage fit of the 1950 instrument.

Known limitations

(i) Wages are deflated by the national PCE index, with no local price adjustment, so cross-CZ comparisons are nominal (Moretti, 2013, shows costs are higher in high-wage cities); (ii) RTI is fixed at 1980 task content, so within-occupation task drift — e.g. the upskilling of clerical work — is not captured; (iii) the data are a decadal CZ panel, not worker-level longitudinal data, so individual transitions are unobserved; (iv) routine intensity and offshorability are strongly positively correlated ($\rho \approx 0.66$), complicating their separation; (v) coverage ends at the 2005 ACS, i.e. before the Great Recession.

4. Empirical strategy

4.1 Defining the high- vs low-routine split

The core regression treats RSH as a continuous regressor, so no split is needed for estimation. For the descriptive U-shape and the binary comparison we define **high-routine CZs** as those with a 1980 routine employment share above the population-weighted national mean of $RSH_{1980} = 0.33$ (our value: 0.3307), reproducing Autor & Dorn’s “above/below grand-mean” split. Robustness: (a) terciles of RSH_{1980} (top vs bottom tercile), and (b) the 80/20 contrast, where $RSH^{P20} = 0.294$ and $RSH^{P80} = 0.365$ (a range of 0.072), which we use to express effect sizes.

4.2 Regression specification — the horse race

We estimate a stacked first-difference model over the three periods 1980–2005, by 2SLS:

$$\Delta SVC_{j,t} = \delta_t + \beta_1 RSH_{j,t0} + \beta_2 \Delta W90_{j,t} + \beta_3 \Delta CollLS_{j,t} + X'_{j,t0} \gamma + \alpha_s + \varepsilon_{j,t}$$

- ΔSVC — decadal change in the noncollege service-employment share (*d_shoccl1_service_nc*).
- RSH_{t0} — start-of-period routine employment share (**Story A: technological push**).
- $\Delta W90$ — change in the 90th-percentile weekly wage; $\Delta CollLS$ — change in college labour supply (**Story B: demand pull — income and substitution effects**).
- X — start-of-period controls (relative skill supply, immigrant share, manufacturing share, unemployment, female employment, elderly share, minimum-wage bite, offshorability); δ_t period and α_s state dummies; SEs clustered on state; weighted by start-of-period CZ population share.

Identification. Because the start-of-period routine share may respond to contemporaneous demand shocks, we instrument RSH_{t0} with the 1950 industry-mix prediction (*R33a_50_**) interacted with period dummies — a measure fixed three decades before the sample, isolating the quasi-fixed component RSH^* (Autor & Dorn, eq. 22). *Predicted signs and adjudication:* $\beta_1 > 0$ under technological push; $\beta_2, \beta_3 > 0$ under demand pull. If β_1 stays large and significant while the demand-pull terms attenuate toward zero, technology dominates; the reverse pattern would favour demand pull. We report the full staircase: (i) RSH only, (ii) demand-pull proxies only, (iii) both, (iv) both + full controls + IV.

Instrument validity. The instrument is strong: in Autor & Dorn’s first stage the 1950 industry-mix measure predicts the contemporaneous routine share with t -ratios above six in every decade (and a coefficient near unity in 1950, declining as initial conditions fade), so weak-instrument bias is not a concern. The exclusion restriction is that a CZ’s 1950 industrial composition affects service-employment growth four to five decades later *only* through its persistent imprint on routine specialisation, not through any direct channel correlated with 1980–2005 demand shocks. We will not claim this holds by assumption: following the package we exclude own-state employment when building the instrument (removing mechanical correlation), and we report whether the second-stage estimate is stable when contemporaneous manufacturing decline and unemployment are added as controls — if the routine-share coefficient is unmoved, the case that the instrument is not simply proxying for a deindustrialisation shock is strengthened.

Feasibility. Preliminary replication runs on the package reproduce Autor & Dorn’s magnitudes (the routine-share coefficient is ≈ 0.11 in the full instrumented specification and remains significant), confirming the pipeline is sound and the horse race is estimable. Table 1 previews these preliminary (weighted OLS) estimates.

Table 1 — Preliminary horse-race estimates (illustrative, weighted OLS)

Dep. var.: ΔSVC (noncollege)	(1)	(2)	(3)	(4)
RSH_{t_0}	0.124***	—	0.096***	0.173***
$\Delta W90$ (top wage)	—	0.048***	0.037**	0.012
$\Delta Colls$	—	—	—	-0.110***
Offshorability	—	—	—	-0.005
N	2,166	2,166	2,166	2,166

Notes: Stacked 1980–2005 (three decadal differences). Weighted by start-of-period CZ population share; standard errors clustered on state. *** $p < 0.01$, ** $p < 0.05$. These are illustrative OLS estimates demonstrating feasibility; the final specification instruments RSH with the 1950 industry-mix measure. The income proxy attenuates and college labour supply turns negative once routine share enters — the conditional pattern discussed in §4.4.

4.3 Two visualisations

- **Output 1 (descriptive) — the U-shape.** Employment- and wage-growth by 1980 occupational-skill percentile, 1980–2005, overlaying high- vs low-routine CZ groups (from *dwg-byperc* and *RTI-bypctile*). Expected result: a deeper, more pronounced U in high-routine CZs.
- **Output 2 (analytical) — the horse-race scatter.** A 722-CZ scatter of ΔSVC against RSH_{1980} with a fitted regression line and a robust-SE band (Panel A), and against top-/college-wage growth (Panel B). Comparing the two fitted slopes visualises which mechanism the data favour; the regression in §4.2 is its formal counterpart with robust standard errors, satisfying the Track-B requirement for a non-trivial analytical output (scatter with fitted line / bivariate regression with robust SEs).

4.4 Treating the RTI-wage relationship across CZs

We treat this relationship explicitly, because its sign is exactly what adjudicates between the mechanisms. In the replication data the *unconditional* cross-CZ correlation between RSH_{1980} and local top-/college-wage growth is **positive and modest** ($\rho \approx 0.29$ at the decadal level; ≈ 0.39 for college labour supply, population-weighted) — not negative. The two candidate drivers are *entangled* across space rather than opposed: routine-intensive CZs also enjoyed faster top-wage growth. This is why a naive bivariate scatter cannot separate them, and precisely why we (a) instrument RSH with predetermined 1950 industry structure (purging the contemporaneous shocks that move both routine share and service demand) and (b) enter the demand-pull proxies directly in the horse race.

The discriminating evidence is *conditional*, not unconditional: once RSH is controlled for, the income proxy attenuates toward zero and the substitution proxy (college labour-supply growth) enters service-employment growth with a **negative** sign (≈ -0.11 in our preliminary runs, matching Autor & Dorn’s Table 6, Panel C). A genuine demand-pull mechanism would predict the opposite. We therefore treat the RTI-wage relationship as the hinge of the test: its positive unconditional sign creates the identification problem, and its conditional disappearance is the evidence that adjudicates in favour of technological push.

4.5 Robustness and falsification checks

Because the rubric rewards an analytical output that is coherent with the research question rather than merely descriptive, we plan a compact battery of checks around the central horse race:

- **Pre-period falsification.** The technological-push story predicts that the routine-share / service-growth link should be a feature of the computer era, not earlier. In Autor & Dorn’s own decade-by-decade estimates the relationship is *negative* in 1950–1970 and absent in the 1970s before turning strongly positive from the 1980s. We replicate this timing as a placebo: a coefficient that is flat or negative before 1980 and positive afterwards is hard to reconcile with a time-invariant demand-pull confounder.
- **Cutoff sensitivity.** Re-estimate the split at the grand mean, at the median, and across terciles, and re-express the continuous effect over the 80/20 range, to confirm the result does not hinge on one threshold.
- **Disaggregation.** Re-run the service-employment regression separately by sex and across the nine service sub-occupations (food, cleaning/gardening, health support, child care, personal appearance, security, recreation, housekeeping, miscellaneous), checking that the routine-share effect is broad-based rather than driven by a single category.
- **Confounder controls.** Add the offshorability index and contemporaneous manufacturing decline; the technology interpretation requires the routine-share coefficient to survive their inclusion, as it does in our preliminary runs.

5. Expected contribution and limitations

What a successful project would establish. (i) A faithful replication of Autor & Dorn’s instrumented routine-share / service-employment relationship; (ii) the conditional (in)significance of demand-pull proxies in the same design; and (iii) a clear, visual adjudication between the two mechanisms on identical data — a contribution because the existing literature tests each mechanism separately.

What it would not establish. It would not recover a causal local-multiplier estimate in the spirit of Moretti (our design lacks a tradable-employment shift-share instrument and is not built to estimate the multiplier); it cannot speak to within-occupation task dynamics, to post-2005 or recession-era patterns, or to *real* (cost-of-living-adjusted) welfare, since wages are nationally deflated.

Threats and how they are handled. Confounding from the positive routine-wage entanglement is addressed by the 1950 instrument plus the horse race; the offshoring confound ($\rho \approx 0.66$ with routine intensity) is handled by including offshorability as a control, which Autor & Dorn show turns insignificant once routine share enters; and the clerical anomaly (falling employment but rising wages in routine-intensive CZs) is acknowledged as within-occupation selection/upskilling and excluded from any causal claim.

Why the answer matters. The two stories carry different policy implications. If service growth is technological push, it reflects workers displaced into lower-paid manual jobs because the middle of the distribution was hollowed out — a case for retraining and wage support. If it is demand pull, the same growth reflects healthy local multipliers from high-productivity employment — a case for attracting tradable, high-skill industry. Because both stories are observationally similar in the aggregate, distinguishing them on consistent micro-geographic data is not a purely academic exercise; it changes which lever a policymaker should reach for. A modest, well-identified horse race on the canonical dataset is a defensible first step toward that distinction, and a natural foundation for a thesis that could later add the tradable-sector instrument needed for a fully causal multiplier estimate.

Acknowledgement of AI use

In preparing this proposal we used a large-language-model assistant (Claude) to inspect the .dta files, sanity-check summary statistics and candidate regression specifications, and to refine the writing. All modelling choices, interpretations, and the final notebook code are our own, and we can explain every line of the submitted code.

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